

Parametric Equations & Modeling



Evaluating parametric equations

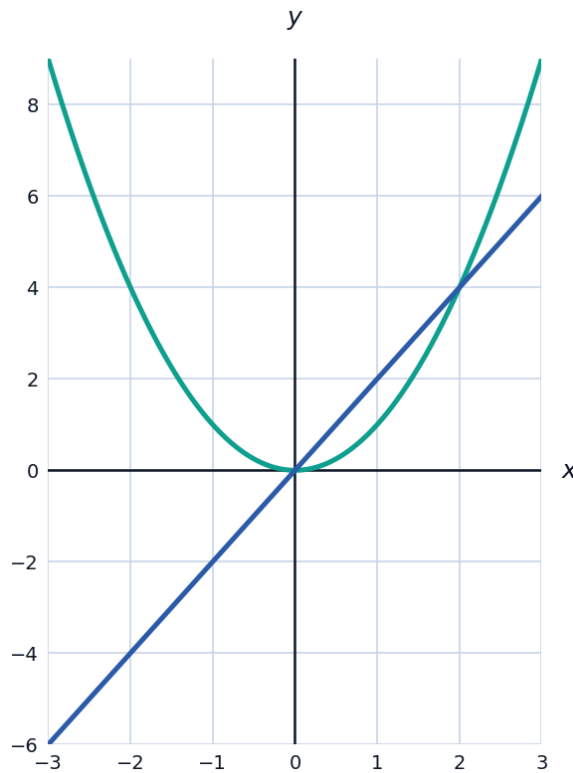
- 1 Given $x(t) = t^2 - 1$ and $y(t) = 2t + 3$, find the point on the curve at $t = 2$.
 - 2 Given $x(t) = 3\cos t$ and $y(t) = 3\sin t$, find the point at $t = \frac{\pi}{3}$, and identify the shape traced by the curve.
-

Eliminating the parameter

- 3 Eliminate the parameter from $x(t) = t + 1$, $y(t) = t^2 - 2$ to find the rectangular equation, and identify the curve.
 - 4 Eliminate the parameter from $x(t) = 2t - 3$, $y(t) = 4t + 1$ to find the rectangular equation.
-

Visualizing a parametric curve's components

- 5 The graph shows the two component functions $x(t) = t^2$ (upper curve) and $y(t) = 2t$ (lower, straight line) for the parametric curve $x(t) = t^2$, $y(t) = 2t$, plotted separately against t on the horizontal axis.



- a Eliminate the parameter t to find the rectangular equation relating x and y for this curve.
- b Use your rectangular equation to describe the shape of the actual parametric curve (not the component graphs shown).

Parametric equations for lines and conics

- 6 Write parametric equations for the line segment from $(1, 2)$ to $(5, 10)$, using $t \in [0, 1]$.
- 7 Write parametric equations for an ellipse with semi-major axis 5 (horizontal) and semi-minor axis 3 (vertical), centered at the origin.

Modeling projectile motion parametrically

- 8 A ball is launched with initial speed 40 m/s at an angle of 30° above horizontal. Its position is modeled by $x(t) = 40\cos(30^\circ)t$ and $y(t) = 40\sin(30^\circ)t - 4.9t^2$. Find the ball's position at $t = 2$ seconds, to one decimal place.
- 9 For the same projectile, find the time at which the ball lands (returns to $y = 0$).

Comparing parametric speed and direction

- 10** Two objects follow the same path $y = x^2$, but object A uses $x(t) = t$, $y(t) = t^2$ while object B uses $x(t) = 2t$, $y(t) = 4t^2$. Explain why they trace the same curve but do not move through it at the same rate.
-

Restricting the parameter's domain

- 11** The parametric equations $x(t) = \cos t$, $y(t) = \sin t$ trace the full unit circle for $t \in [0, 2\pi)$. Describe the portion of the circle traced when the domain is restricted to $t \in [0, \frac{\pi}{2}]$.
- 12** Explain why restricting t in $x(t) = t^2$, $y(t) = t$ to $t \geq 0$ changes the rectangular relation from $x = y^2$ (a full sideways parabola) to only its upper half.
-

Parametric equations for circular motion with a phase shift

- 13** A point moves along a circle of radius 4 according to $x(t) = 4\cos\left(t + \frac{\pi}{4}\right)$, $y(t) = 4\sin\left(t + \frac{\pi}{4}\right)$. Find its position at $t = 0$, and explain the effect of the $\frac{\pi}{4}$ phase shift compared to $x(t) = 4\cos t$, $y(t) = 4\sin t$.